BBA · Applied Mathematics for Business · 2026–2027

Unit 2 Practice Papers

One-Variable Calculus · Topics 2.1 – 2.5

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Mock A, Mock B	Mixed exams, 60 minutes each
At the end	Answer sheets with worked solutions

How to use these papers

1. Do each paper on paper, timed, with only the formula sheet next to you.
2. Write the formula you use before you put numbers in. Finish every application question with one sentence in context, with units.
3. Mark your work with the answer sheets at the back. For every mistake, write one line on *why* it happened.
4. Do the two mock exams last, under real exam conditions (60 minutes, no notes).

Formula sheet

This is the kind of sheet you can expect next to the exam: definitions and rules only, no worked examples. Learn where each formula is so you can find it in seconds.

2.1 · Domains

$\frac{1}{u}$: $u \neq 0$ $\quad \sqrt{u}$: $u \geq 0$ $\quad \ln u$: $u > 0$

f	$D(f)$	$R(f)$
x^2	$\mathbb{R}$	$[0,+\infty)$
$\sqrt{x}$	$[0,+\infty)$	$[0,+\infty)$
$\frac{1}{x}$	$\mathbb{R}\setminus\{0\}$	$\mathbb{R}\setminus\{0\}$
e^x	$\mathbb{R}$	$(0,+\infty)$
$\ln x$	$(0,+\infty)$	$\mathbb{R}$
$\lvert x\rvert$	$\mathbb{R}$	$[0,+\infty)$
$\lfloor x\rfloor$	$\mathbb{R}$	$\mathbb{Z}$

$(f \circ g)(x) = f\big(g(x)\big)$ $\quad$ (inside function: g)

2.1 · Limits

$\lim_{x\to a} f(x) = L \iff \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L$

$\frac{k}{\pm\infty} = 0$ $\quad \frac{k}{0^+} = +\infty$ $\quad \frac{k}{0^-} = -\infty$ $\quad (k > 0)$

$e^{+\infty} = +\infty$ $\quad e^{-\infty} = 0$ $\quad \ln(+\infty) = +\infty$ $\quad \ln(0^+) = -\infty$

Vertical asymptote $x = a$: a one-sided limit is $\pm\infty$.
Horizontal asymptote $y = L$: $\lim_{x\to\pm\infty} f(x) = L$.

2.1 · Continuity at $x = a$

1. $f(a)$ exists $\quad$ 2. $\lim_{x\to a} f(x)$ exists $\quad$ 3. they are equal

Removable: limit exists $\neq f(a)$
Jump: finite one-sided limits, different
Infinite: a one-sided limit is $\pm\infty$

Business

$I(q) = q \cdot p(q)$ $\quad \pi(q) = I(q) - C(q)$
Break-even: $\pi(q) = 0$ $\quad$ Average cost: $A(q) = \frac{C(q)}{q}$

Marginal = derivative $\quad$ Units of f': $\frac{\text{units of } f}{\text{units of } x}$

2.2 · Derivatives

$(a + bx)' = b$ $\quad (x^n)' = n\,x^{n-1}$

$(e^x)' = e^x$ $\quad (\ln x)' = \frac{1}{x}$

$(\sqrt{x})' = \frac{1}{2\sqrt{x}}$ $\quad \left(\frac{1}{x}\right)' = -\frac{1}{x^2}$

$(f\,g)' = f'g + f\,g'$

$\left(\frac{f}{g}\right)' = \frac{f'g - f\,g'}{g^2}$

$\big(f(g(x))\big)' = f'\big(g(x)\big)\cdot g'(x)$

$\big(e^u\big)' = e^u u'$ $\quad (\ln u)' = \frac{u'}{u}$ $\quad (u^n)' = n\,u^{n-1}u'$

Tangent: $L(x) = f(a) + f'(a)(x - a)$

Average rate of change: $\frac{f(b) - f(a)}{b - a}$

Differentiable $\Rightarrow$ continuous (not the reverse: $|x|$ at 0)

2.3 · Analysis & optimization

Stationary point: $f'(x^*) = 0$
$f' > 0$ increasing, $f' < 0$ decreasing
$f''(x^*) > 0$ local min, $f''(x^*) < 0$ local max,
$f''(x^*) = 0$ inconclusive
Inflection: $f''(a) = 0$ *and* f'' changes sign

Candidates for global max/min: stationary points, endpoints of the (contextual) domain, kinks and discontinuities. Evaluate f at all of them.

2.4 · Taylor

$$P_n(x) = \sum_{i=0}^{n} \frac{f^{(i)}(a)}{i!}(x - a)^i, \qquad P_1(x) = L(x)$$

Coefficient c_k of $(x - a)^k$: $f^{(k)}(a) = k! \cdot c_k$

$2! = 2$ $\quad 3! = 6$ $\quad 4! = 24$ $\quad 5! = 120$

Sign of $f - L$ near a = sign of first term after the linear one, $c(x - a)^m$:
m even: $c > 0$ above (concave up), $c < 0$ below
m odd: crosses the tangent (inflection)

$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$

2.5 · Integration

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \ (n \neq -1) \qquad \int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^{kx}\,dx = \frac{e^{kx}}{k} + C \qquad \int k\,dx = kx + C$$

$$\int_a^b f(x)\,dx = F(b) - F(a) \quad \text{(signed area)}$$

$$\int_a^b f = \int_a^c f + \int_c^b f \qquad \int_b^a f = -\int_a^b f$$

$$C(b) - C(a) = \int_a^b C'(x)\,dx \qquad C(x) = C(0) + \int_0^x C'(t)\,dt$$

Average value: $\dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$

Density: $f \geq 0$, $\displaystyle\int_{-\infty}^{+\infty} f = 1$

Paper 1 · Functions, limits & continuity (2.1)

Time: 40 minutes **Total:** 40 marks Formula sheet allowed

1. Find the domain and the range of each function. [6]

(a) $f(x) = \sqrt{4 - 2x}$

(b) $g(x) = \ln(x - 3)$

(c) $h(x) = \dfrac{1}{x^2 - 9}$ (domain only)

(d) $k(x) = e^x + 1$

2. Let $f(x) = x^2 + 1$ and $g(x) = \ln x$. [5]

(a) Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

(b) Give the domain and range of both compositions.

3. Calculate the following limits, or explain why they do not exist. [8]

(a) $\displaystyle\lim_{x \to +\infty} \left(3 - \frac{5}{x}\right)$

(b) $\displaystyle\lim_{x \to -\infty} (e^x + 2)$

(c) $\displaystyle\lim_{x \to 2^-} \frac{1}{x - 2}$, $\displaystyle\lim_{x \to 2^+} \frac{1}{x - 2}$ and $\displaystyle\lim_{x \to 2} \frac{1}{x - 2}$

(d) $\displaystyle\lim_{x \to 0^+} \ln x$

(e) $\displaystyle\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$

4. Let

$$f(x) = \begin{cases} 2x + 1 & \text{if } x < 1 \\ x^2 + a & \text{if } x \geq 1 \end{cases}$$

[5]

(a) Find the value of a that makes f continuous on $\mathbb{R}$.

(b) Classify the discontinuity at $x = 1$ when $a = 5$.

5. Study the continuity of

$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 1 & \text{if } x = 0 \\ e^{-x} & \text{if } 0 < x \leq 2 \\ \dfrac{1}{x - 2} & \text{if } x > 2 \end{cases}$$

Classify every discontinuity and give all asymptotes. Do not use a graph as justification. [8]

6. A firm sells q units at a price of $p(q) = 60 - 0.5q$ euros per unit. Its total cost is $C(q) = 200 + 10q$ euros. [8]

(a) Give the contextual domain of $p(q)$.

(b) Find the income function $I(q)$ and the profit function $\pi(q)$.

(c) Find the break-even quantities and say which one is the break-even point.

Paper 2 · Derivatives (2.2)

Time: 45 minutes **Total:** 45 marks Formula sheet allowed

1. Differentiate. Simplify where possible. [16]

(a) $f(x) = 3x^4 - 2x^2 + 7x - 5$

(b) $f(x) = \sqrt{x} + \dfrac{1}{x^2}$

(c) $f(x) = x^3 e^{2x}$

(d) $f(x) = \ln(x^2 + 4)$

(e) $f(x) = \dfrac{x+1}{x-1}$

(f) $f(x) = e^{-x^2}$

(g) $f(x) = (2x-3)^5$

(h) $f(x) = x \ln x$

2. Find the tangent line to [6]

(a) $f(x) = x^3 - 2x$ at $x = 1$;

(b) $g(x) = \ln(2x)$ at $x = \frac{1}{2}$.

3. Daily revenue, in thousands of euros, is $R(t) = 0.02t^3 - 0.9t^2 + 12t + 50$, where t is the number of days since launch. [9]

(a) Calculate the average rate of change of revenue between day 10 and day 12.

(b) Find $R'(t)$ and calculate $R'(5)$. Interpret it with units.

(c) Use the tangent line at $t = 5$ to estimate $R(5.5)$, and compare it with the exact value.

4. A company has total cost $C(x) = 500 + 20x + 0.05x^2$ euros and average cost $A(x) = \dfrac{C(x)}{x}$. [7]

(a) Calculate $C'(50)$ and $A'(50)$ and explain the difference in meaning.

(b) Estimate $C(51)$ with the tangent line at $x = 50$.

5. Demand for a product is $q(p) = 1000e^{-0.05p}$, where p is the price in euros. Revenue is $I(p) = p\,q(p)$. [5]

(a) Find $q'(p)$ and $I'(p)$.

(b) Calculate $q'(10)$ and $I'(10)$ and interpret both.

6. Is $f(x) = |x - 2|$ continuous at $x = 2$? Is it differentiable there? Justify. [2]

Paper 3 · Analysis & optimization (2.3)

Time: 45 minutes **Total:** 45 marks Formula sheet allowed

1. Let $f(x) = x^3 - 6x^2 + 9x + 1$. [10]

(a) Find the intervals where f is increasing and decreasing.

(b) Find and classify all local extrema.

(c) Find the inflection point and the intervals of concavity.

2. Find the global maximum and minimum of the same $f(x) = x^3 - 6x^2 + 9x + 1$ on $[0, 4]$. [6]

3. Profit is $\pi(q) = -0.5q^2 + 50q - 200$ euros. Find the production level that maximizes profit and the maximum profit. Justify that it is a maximum. [5]

4. Monthly profit, in thousands of euros, is $P(x) = 100 - 120e^{-0.1x} - 4x$, where x is the advertising budget in thousands of euros and $0 \le x \le 40$. [10]

(a) Find all stationary points in the contextual domain.

(b) Classify them with the second derivative.

(c) Find the global maximum and minimum and interpret them in context.

5. Let $f(x) = x\,e^{-x}$. Find its local extrema and its inflection point. [6]

6. Profit, with x in hundreds of units, is

$$P(x) = \begin{cases} 3x & 0 \le x \le 2 \\ -x^2 + 6x - 2 & 2 < x \le 5 \end{cases}$$

[6]

(a) Show that P is continuous at $x = 2$. Is it differentiable there?

(b) Find the production level that maximizes profit.

7. True or false? Justify. "If $f''(a) = 0$, then $x = a$ is an inflection point of f." [2]

Paper 4 · Taylor approximation (2.4)

Time: 40 minutes **Total:** 40 marks Formula sheet allowed

1. Find the third-order Taylor polynomial of $f(x) = e^{2x}$ around $x = 0$. [5]

2. Let $f(x) = \sqrt{x}$. [7]

(a) Find $P_2(x)$ around $x = 4$.

(b) Use it to estimate $\sqrt{4.1}$.

3. Find $P_3(x)$ of $f(x) = \ln(1 + x)$ around $x = 0$. [5]

4. The fifth-order Taylor polynomial of f around $a = 1$ is

$$P_5(x) = 3 - 2(x - 1) + 4(x - 1)^4 - (x - 1)^5.$$

[7]

(a) Find $f(1)$, $f'(1)$, $f''(1)$, $f'''(1)$, $f^{(4)}(1)$ and $f^{(5)}(1)$.

(b) Write the tangent line at $x = 1$ and sketch f near $x = 1$. Justify.

5. A function h satisfies $h(2) = 1$, $h'(2) = h''(2) = h'''(2) = 0$ and $h^{(4)}(2) = -48$. [6]

(a) Find $P_4(x)$ around $x = 2$.

(b) What kind of point is $x = 2$? Sketch h nearby.

6. The Taylor polynomial of g around $a = -1$ is $P_3(x) = 5 + (x + 1) + 2(x + 1)^3$. Describe the graph of g near $x = -1$ relative to its tangent line. [5]

7. Monthly profit is $\pi(t) = e^t - \dfrac{t^2}{2} - t$, in millions of euros. Find $P_3(t)$ around $t = 0$ and say what happens to profit around $t = 0$. [5]

Paper 5 · Integration (2.5)

Time: 45 minutes **Total:** 45 marks Formula sheet allowed

1. Find these indefinite integrals. [9]

(a) $\int (6x^2 - 4x + 3)\,dx$

(b) $\int \left(\frac{1}{x} + \sqrt{x}\right) dx$

(c) $\int e^{3x}\,dx$

(d) $\int \frac{5}{x^3}\,dx$

(e) $\int 2e^{-0.5x}\,dx$

2. Calculate. [7]

(a) $\int_0^2 (3x^2 + 1)\,dx$

(b) $\int_1^e \frac{2}{x}\,dx$

(c) $\int_0^{\ln 2} e^x\,dx$

(d) $\int_{-1}^1 x^3\,dx$

3. Find k such that $\int_0^3 (kx + 2)\,dx = 15$. [4]

4. Marginal cost is $C'(q) = \frac{4}{\sqrt{q}}$ euros per unit, and producing 25 units costs €300. Find $C(q)$ and the cost of producing 100 units. [6]

5. Marginal revenue is $R'(x) = 50 - 0.2x$ euros per unit, with $R(0) = 0$. [5]

(a) Find the revenue function $R(x)$.

(b) By how much does revenue increase when sales rise from 100 to 200 units?

6. [6]

(a) Calculate $\int_{-1}^2 x^2\,dx$.

(b) Calculate $\int_0^3 (x - 1)\,dx$ and the area between $y = x - 1$ and the x-axis on $[0, 3]$. Why are they different?

7. Find the average value of $f(x) = 3x^2$ on $[0, 2]$. [3]

8. Let $f(x) = kx$ for $0 \le x \le 2$ and $f(x) = 0$ otherwise. [3]

(a) Find k so that f is a density function.

(b) Calculate $\int_0^1 f(x)\,dx$.

9. Calculate $\int_0^3 \lfloor x \rfloor\,dx$. [2]

Mock exam A · Topics 2.1 – 2.5

Time: 60 minutes **Total:** 60 marks Formula sheet allowed No notes

1. Let

$$f(x) = \begin{cases} e^x & \text{if } x \le 0 \\ x^2 + b & \text{if } x > 0 \end{cases}$$

[10]

(a) Find b so that f is continuous on $\mathbb{R}$.

(b) With that value of b, is f differentiable at $x = 0$? Justify.

(c) Find $\lim_{x \to -\infty} f(x)$ and $\lim_{x \to +\infty} f(x)$. Which asymptotes does f have?

2. Let $f(x) = x^2 e^{-x}$. [10]

(a) Find $f'(x)$ and simplify.

(b) Find the tangent line at $x = 1$.

(c) Find the stationary points and classify them.

3. Units sold depend on price: $q = 200 - 4p$. Total cost is $C(q) = 500 + 10q$ euros. [14]

(a) Write profit $\pi(p)$ as a function of price, and give its contextual domain.

(b) Find the price that maximizes profit, the units sold and the maximum profit.

(c) Show that it is the global maximum on the contextual domain.

4. Let $f(x) = e^x - x$. [8]

(a) Find $P_3(x)$ around $x = 0$.

(b) Use it to describe f near $x = 0$. What kind of point is $x = 0$?

5. Marginal cost is $C'(x) = 6 + 0.02x$ euros per unit, and the fixed cost is €1,000. [10]

(a) Find the total cost function $C(x)$.

(b) By how much does total cost increase when production rises from 100 to 200 units?

(c) Find the average cost per unit at $x = 100$.

6. True or false? Justify each in one line. [8]

(a) If f is continuous at a, then f is differentiable at a.

(b) The global maximum of a function on $[a, b]$ is always at a point where $f' = 0$.

(c) $\int_a^b f(x)\,dx$ is always the area between f and the x-axis.

(d) The tangent line at a is the first-order Taylor polynomial at a.

Mock exam B · Topics 2.1 – 2.5

Time: 60 minutes **Total:** 60 marks Formula sheet allowed No notes

1. Let $f(x) = \ln(5 - x) + \sqrt{x + 1}$. [6]

(a) Find the domain of f.

(b) Calculate $\lim_{x \to 5^-} f(x)$. What does it tell you about the graph?

2. Let $g(x) = \dfrac{\ln x}{x}$ for $x > 0$. [10]

(a) Find $g'(x)$ using the quotient rule.

(b) Find where g is increasing and decreasing, and its maximum.

3. The number of subscribers of an app is $N(t) = \dfrac{5000}{1 + t}$, with t in months. [8]

(a) Find $N'(t)$ and calculate $N'(4)$. Interpret it with units.

(b) Use the tangent line at $t = 4$ to estimate $N(4.5)$.

4. The Taylor polynomial of f around $a = 3$ is $P_3(x) = 2 - (x - 3) + \dfrac{1}{2}(x - 3)^3$. [8]

(a) Find $f(3)$, $f'(3)$, $f''(3)$ and $f'''(3)$.

(b) Sketch f near $x = 3$ and justify.

5. Calculate $\displaystyle\int_1^4 \left(3\sqrt{x} - \frac{2}{x}\right) dx$. Give the exact value and a decimal. [8]

6. Find the area of the region enclosed by $y = x^2$ and $y = 2x$. [10]

7. Daily profit, in thousands of euros, is $P(t) = 12 + 6t - t^2$ for $0 \le t \le 6$ hours of opening. [10]

(a) Find the time at which profit is highest and the maximum profit.

(b) Find the average value of P on $[0, 6]$.

Answer sheets

Worked solutions. Check your method, not only the final number.

Paper 1 · Functions, limits & continuity

1. (a) $4 - 2x \geq 0 \Rightarrow x \leq 2$. $D = (-\infty, 2]$, $R = [0, +\infty)$.

(b) $x - 3 > 0$. $D = (3, +\infty)$, $R = \mathbb{R}$.

(c) $x^2 - 9 \neq 0 \Rightarrow x \neq \pm 3$. $D = \mathbb{R} \setminus \{-3, 3\}$.

(d) $D = \mathbb{R}$. Since $e^x > 0$, $R = (1, +\infty)$.

2. (a) $(f \circ g)(x) = f(\ln x) = (\ln x)^2 + 1$. $(g \circ f)(x) = g(x^2 + 1) = \ln(x^2 + 1)$.

(b) $f \circ g$: needs $x > 0$, so $D = (0, +\infty)$; $(\ln x)^2 \geq 0$ so $R = [1, +\infty)$.
$g \circ f$: $x^2 + 1 > 0$ always, so $D = \mathbb{R}$; $x^2 + 1 \geq 1$ so $R = [0, +\infty)$.

3. (a) $3 - \dfrac{5}{+\infty} = 3 - 0 = 3$.

(b) $e^{-\infty} + 2 = 0 + 2 = 2$ (horizontal asymptote $y = 2$ on the left).

(c) $x \to 2^-$: denominator $\to 0^-$, limit $-\infty$. $x \to 2^+$: limit $+\infty$. The one-sided limits are different, so $\lim_{x \to 2}$ has no single signed value (the course writes ∞). Vertical asymptote $x = 2$.

(d) $-\infty$.

(e) Factor: $\dfrac{(x-3)(x+3)}{x-3} = x + 3 \to 6$. (The limit ignores the value at $x = 3$.)

4. (a) Left: $\lim_{x \to 1^-}(2x + 1) = 3$. Right and value: $f(1) = 1 + a$. $3 = 1 + a \Rightarrow a = 2$.

(b) With $a = 5$: left limit 3, right limit 6, both finite and different: **jump discontinuity**.

5. Each piece is continuous on its own interval, so check $x = 0$ and $x = 2$.
$x = 0$: left $\to 0^2 = 0$, right $\to e^0 = 1$, $f(0) = 1$. One-sided limits differ: **jump discontinuity**.
$x = 2$: left $\to e^{-2}$, right $\to \dfrac{1}{0^+} = +\infty$: **infinite discontinuity**; vertical asymptote $x = 2$ (from the right).
$x \to +\infty$: $\dfrac{1}{x-2} \to 0$: horizontal asymptote $y = 0$ on the right. $x \to -\infty$: $x^2 \to +\infty$, no asymptote on the left.

6. (a) $p > 0 \iff 60 - 0.5q > 0 \iff q < 120$, and $q \geq 0$: $D_c = [0, 120)$.

(b) $I(q) = q(60 - 0.5q) = 60q - 0.5q^2$. $\pi(q) = 60q - 0.5q^2 - 200 - 10q = -0.5q^2 + 50q - 200$.

(c) $-0.5q^2 + 50q - 200 = 0 \iff q^2 - 100q + 400 = 0 \iff q = \dfrac{100 \pm \sqrt{8400}}{2}$, so $q \approx 4.17$ or $q \approx 95.83$. The break-even point is $q \approx 4.17$ units, where losses turn into profit. After 95.83 units the firm makes losses again.

Paper 2 · Derivatives

1. (a) $12x^3 - 4x + 7$

(b) $\dfrac{1}{2\sqrt{x}} - \dfrac{2}{x^3}$

(c) Product rule: $3x^2e^{2x} + x^3 \cdot 2e^{2x} = x^2e^{2x}(3 + 2x)$

(d) Chain rule: $\dfrac{2x}{x^2 + 4}$

(e) Quotient rule: $\dfrac{1\cdot(x-1)-(x+1)\cdot 1}{(x-1)^2}=\dfrac{-2}{(x-1)^2}$

(f) $-2x\,e^{-x^2}$

(g) $5(2x-3)^4\cdot 2=10(2x-3)^4$

(h) $1\cdot\ln x+x\cdot\dfrac{1}{x}=\ln x+1$

2. (a) $f(1)=-1$, $f'(x)=3x^2-2$, $f'(1)=1$. $L(x)=-1+1\cdot(x-1)=x-2$.

(b) $g(\frac{1}{2})=\ln 1=0$, $g'(x)=\dfrac{2}{2x}=\dfrac{1}{x}$, $g'(\frac{1}{2})=2$. $L(x)=2\left(x-\frac{1}{2}\right)=2x-1$.

3. (a) $R(10)=100$, $R(12)=98.96$. $\dfrac{98.96-100}{12-10}=-0.52$ thousand €/day: on average revenue fell by about €520 per day.

(b) $R'(t)=0.06t^2-1.8t+12$, $R'(5)=1.5-9+12=4.5$. On day 5 revenue is rising at about €4,500 per day.

(c) $R(5)=90$, so $L(t)=90+4.5(t-5)$ and $R(5.5)\approx 92.25$ thousand €. Exact: $R(5.5)=92.1025$. The estimate is €147.50 too high.

4. (a) $C'(x)=20+0.1x$, $C'(50)=25$: one more unit costs about €25.
$A(x)=\dfrac{500}{x}+20+0.05x$, $A'(x)=-\dfrac{500}{x^2}+0.05$, $A'(50)=-0.2+0.05=-0.15$: the cost *per unit* falls by about €0.15.
C' is about total cost, A' about cost per unit.

(b) $C(50)=1625$. $C(51)\approx 1625+25\cdot 1=1650$. (Exact: €1650.05.)

5. (a) $q'(p)=1000e^{-0.05p}\cdot(-0.05)=-50e^{-0.05p}$.
$I'(p)=1000e^{-0.05p}+1000p\cdot(-0.05)e^{-0.05p}=1000e^{-0.05p}(1-0.05p)$.

(b) $q'(10)=-50e^{-0.5}\approx -30.33$: at €10, a €1 price rise loses about 30 units of demand.
$I'(10)=1000e^{-0.5}\cdot 0.5\approx 303.27$: at €10, a €1 price rise still adds about €303 of revenue.

6. Continuous: $\lim\limits_{x\to 2}|x-2|=0=f(2)$. Not differentiable: the slope is -1 on the left and $+1$ on the right, so there is a corner and no unique tangent.

Paper 3 · Analysis & optimization

1. (a) $f'(x)=3x^2-12x+9=3(x-1)(x-3)$. Positive on $(-\infty,1)$ and $(3,+\infty)$: increasing. Negative on $(1,3)$: decreasing.

(b) $f''(x)=6x-12$. $f''(1)=-6<0$: local max $(1,5)$. $f''(3)=6>0$: local min $(3,1)$.

(c) $f''=0$ at $x=2$ and changes sign: inflection point $(2,3)$. Concave down on $(-\infty,2)$, up on $(2,+\infty)$.

2. Candidates: $x=0,1,3,4$. $f(0)=1$, $f(1)=5$, $f(3)=1$, $f(4)=5$. Global max 5 at $x=1$ *and* $x=4$; global min 1 at $x=0$ *and* $x=3$. (An endpoint ties with the stationary point.)

3. $\pi'(q)=-q+50=0\Rightarrow q^*=50$. $\pi''=-1<0$: maximum. $\pi(50)=-1250+2500-200=1050$.

4. (a) $P'(x)=-120e^{-0.1x}(-0.1)-4=12e^{-0.1x}-4=0\Rightarrow e^{-0.1x}=\frac{1}{3}\Rightarrow x=10\ln 3\approx 10.99$.

(b) $P''(x)=-1.2e^{-0.1x}<0$: local maximum.

(c) $P(0)=-20$, $P(10\ln 3)=100-40-43.94\approx 16.06$, $P(40)\approx -62.20$. Spend about €10,990 on advertising for a maximum profit of about €16,060 a month. The global minimum is at $x=40$.

5. $f'(x) = e^{-x} - xe^{-x} = e^{-x}(1-x) = 0 \Rightarrow x = 1$. $f''(x) = e^{-x}(x-2)$, $f''(1) = -e^{-1} < 0$: local max $\left(1, \frac{1}{e}\right)$. $f'' = 0$ at $x = 2$ with a sign change: inflection point $\left(2, \frac{2}{e^2}\right)$.

6. (a) Left: $3 \cdot 2 = 6$. Right: $-4+12-2 = 6$. $P(2) = 6$: continuous. Slopes: left 3, right $-2x+6 \to 2$. Different, so a kink: not differentiable at $x = 2$.

(b) On $(2, 5]$: $P'(x) = -2x + 6 = 0 \Rightarrow x = 3$, $P(3) = 7$. Compare: $P(0) = 0$, $P(2) = 6$, $P(3) = 7$, $P(5) = 3$. Maximum at $x = 3$: produce **300 units** for a profit of 7.

7. False. Example: $f(x) = x^4$ has $f''(0) = 0$, but $f'' = 12x^2 \geq 0$ never changes sign: $x = 0$ is a minimum, not an inflection point.

Paper 4 · Taylor approximation

1. $f^{(n)}(x) = 2^n e^{2x}$, so $f(0) = 1$, $f'(0) = 2$, $f''(0) = 4$, $f'''(0) = 8$.
$P_3(x) = 1 + 2x + \frac{4}{2}x^2 + \frac{8}{6}x^3 = 1 + 2x + 2x^2 + \frac{4}{3}x^3$.

2. (a) $f(4) = 2$, $f'(x) = \frac{1}{2\sqrt{x}}$, $f'(4) = \frac{1}{4}$, $f''(x) = -\frac{1}{4x^{3/2}}$, $f''(4) = -\frac{1}{32}$.
$P_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$.

(b) $\sqrt{4.1} \approx 2 + 0.025 - \frac{0.01}{64} \approx 2.02484$ (true value 2.024846).

3. $f(0) = 0$, $f' = \frac{1}{1+x} \to 1$, $f'' = -\frac{1}{(1+x)^2} \to -1$, $f''' = \frac{2}{(1+x)^3} \to 2$.
$P_3(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$.

4. (a) $f(1) = 3$, $f'(1) = -2$, $f''(1) = f'''(1) = 0$, $f^{(4)}(1) = 4! \cdot 4 = 96$, $f^{(5)}(1) = 5! \cdot (-1) = -120$.

(b) $L(x) = 3 - 2(x-1)$. First term after the linear one: $4(x-1)^4 \geq 0$, an even power with a positive coefficient. So f lies **above** its tangent on both sides: concave up near $x = 1$.

5. (a) $P_4(x) = 1 + \frac{-48}{24}(x-2)^4 = 1 - 2(x-2)^4$.

(b) The tangent is horizontal ($y = 1$) and $f - L \approx -2(x-2)^4 \leq 0$: h lies below its tangent on both sides, so $x = 2$ is a **local maximum**.

6. $L(x) = 5 + (x+1)$. $P_3 - P_1 = 2(x+1)^3$ is an odd power: for $x < -1$ it is negative (below the tangent, concave down), for $x > -1$ positive (above, concave up). $x = -1$ is an **inflection point**.

7. $\pi(0) = 1$, $\pi'(t) = e^t - t - 1 \Rightarrow \pi'(0) = 0$, $\pi''(t) = e^t - 1 \Rightarrow \pi''(0) = 0$.
$\pi'''(t) = e^t \Rightarrow \pi'''(0) = 1$. So $P_3(t) = 1 + \frac{t^3}{6}$ (here P_2 alone would just be 1 and could not show the shape). The tangent is horizontal and the first non-zero term is an odd power: profit has a **stationary inflection point** at $t = 0$. It keeps rising through $t = 0$, flattening for a moment.

Paper 5 · Integration

1. (a) $2x^3 - 2x^2 + 3x + C$

(b) $\ln|x| + \frac{x^{3/2}}{3/2} + C = \ln|x| + \frac{2}{3}x^{3/2} + C$

(c) $\frac{e^{3x}}{3} + C$

(d) $\displaystyle\int 5x^{-3}dx = \frac{5x^{-2}}{-2} + C = -\frac{5}{2x^2} + C$

(e) $\dfrac{2e^{-0.5x}}{-0.5} + C = -4e^{-0.5x} + C$

2. (a) $\Big[x^3 + x\Big]_0^2 = 10$

(b) $\Big[2\ln x\Big]_1^e = 2 - 0 = 2$

(c) $\Big[e^x\Big]_0^{\ln 2} = 2 - 1 = 1$

(d) $\Big[\dfrac{x^4}{4}\Big]_{-1}^1 = \dfrac{1}{4} - \dfrac{1}{4} = 0$ (the areas above and below cancel)

3. $\Big[\dfrac{kx^2}{2} + 2x\Big]_0^3 = \dfrac{9k}{2} + 6 = 15 \Rightarrow k = 2.$

4. $C(q) = \displaystyle\int 4q^{-1/2}dq = \frac{4q^{1/2}}{1/2} + K = 8\sqrt{q} + K.$ $C(25) = 40 + K = 300 \Rightarrow K = 260.$
$C(q) = 8\sqrt{q} + 260,$ $C(100) = 80 + 260 = 340.$

5. (a) $R(x) = 50x - 0.1x^2 + K$, and $R(0) = 0$ gives $K = 0$: $R(x) = 50x - 0.1x^2$.

(b) $\displaystyle\int_{100}^{200} (50 - 0.2x)\,dx = R(200) - R(100) = 6000 - 4000 = 2000.$

6. (a) $\Big[\dfrac{x^3}{3}\Big]_{-1}^2 = \dfrac{8}{3} + \dfrac{1}{3} = 3.$

(b) Integral: $\Big[\dfrac{x^2}{2} - x\Big]_0^3 = \dfrac{9}{2} - 3 = 1.5.$
Area: below the axis on $[0,1]$ (triangle 0.5), above on $[1,3]$ (triangle 2), total **2.5**. The integral counts the part below the axis as negative: $2 - 0.5 = 1.5$.

7. $\dfrac{1}{2-0}\displaystyle\int_0^2 3x^2dx = \frac{1}{2}\cdot 8 = 4.$

8. (a) $\displaystyle\int_0^2 kx\,dx = 2k = 1 \Rightarrow k = \frac{1}{2}$ (and $f \geq 0$).

(b) $\displaystyle\int_0^1 \frac{x}{2}\,dx = \frac{1}{4}.$

9. Rectangles: height 0 on $[0,1)$, 1 on $[1,2)$, 2 on $[2,3)$: $0 + 1 + 2 = 3.$

Mock exam A

1. (a) Left: $e^0 = 1$. Right: $0 + b$. Continuous $\iff b = 1$.

(b) Slopes: left $(e^x)' = e^x \to 1$, right $(x^2+1)' = 2x \to 0$. Different: **not differentiable** at 0 (a kink), even though it is continuous.

(c) $x \to -\infty$: $e^x \to 0$, horizontal asymptote $y = 0$ on the left. $x \to +\infty$: $x^2 + 1 \to +\infty$, no asymptote on the right.

2. (a) $f'(x) = 2xe^{-x} - x^2e^{-x} = x\,e^{-x}(2 - x).$

(b) $f(1) = \dfrac{1}{e}$, $f'(1) = \dfrac{1}{e}$: $L(x) = \dfrac{1}{e} + \dfrac{1}{e}(x-1) = \dfrac{x}{e}.$

(c) $f' = 0 \Rightarrow x = 0$ or $x = 2$. Sign of f': $-$ on $(-\infty, 0)$, $+$ on $(0, 2)$, $-$ on $(2, +\infty)$. Local min $(0, 0)$, local max $\left(2, \dfrac{4}{e^2}\right)$.

3. (a) $\pi(p) = p(200 - 4p) - 500 - 10(200 - 4p) = -4p^2 + 240p - 2500$. We need $q \geq 0$ and $p \geq 0$: $D_c = [0, 50]$.

(b) $\pi'(p) = -8p + 240 = 0 \Rightarrow p = 30$. Then $q = 200 - 120 = 80$ units and $\pi(30) = -3600 + 7200 - 2500 = 1100$.

(c) $\pi'' = -8 < 0$, and at the endpoints $\pi(0) = -2500$, $\pi(50) = -500$, both below 1100. So $p = 30$ is the global maximum.

4. (a) $f(0) = 1$, $f' = e^x - 1 \to 0$, $f'' = e^x \to 1$, $f''' = e^x \to 1$. $P_3(x) = 1 + \dfrac{x^2}{2} + \dfrac{x^3}{6}$.

(b) Tangent: $y = 1$ (horizontal). The first term after it is $\dfrac{x^2}{2} \geq 0$, an even power with a positive coefficient: f is above its tangent on both sides. $x = 0$ is a **local minimum**.

5. (a) $C(x) = 1000 + \displaystyle\int_0^x (6 + 0.02t)\, dt = 1000 + 6x + 0.01x^2$.

(b) $\displaystyle\int_{100}^{200} (6 + 0.02x)\, dx = \Big[6x + 0.01x^2\Big]_{100}^{200} = 1600 - 700 = 900$.

(c) $C(100) = 1000 + 600 + 100 = 1700$, $A(100) = \dfrac{1700}{100} = 17$ per unit.

6. (a) False: $|x|$ is continuous at 0 but not differentiable.

(b) False: it can be at an endpoint or a kink (see Paper 3, Q2 and Q6).

(c) False: parts below the x-axis count as negative (signed area).

(d) True: $P_1(x) = f(a) + f'(a)(x - a) = L(x)$.

Mock exam B

1. (a) $\ln(5 - x)$ needs $x < 5$; $\sqrt{x + 1}$ needs $x \geq -1$. $D = [-1, 5)$.

(b) $x \to 5^-$: $5 - x \to 0^+$, so $\ln(5 - x) \to -\infty$ while $\sqrt{x + 1} \to \sqrt{6}$. The limit is $-\infty$: vertical asymptote $x = 5$ (from the left).

2. (a) $g'(x) = \dfrac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2} = \dfrac{1 - \ln x}{x^2}$.

(b) $x^2 > 0$, so the sign is that of $1 - \ln x$: positive for $0 < x < e$ (increasing), negative for $x > e$ (decreasing). Maximum at $x = e$: $g(e) = \dfrac{1}{e}$.

3. (a) $N(t) = 5000(1 + t)^{-1}$, $N'(t) = -\dfrac{5000}{(1 + t)^2}$, $N'(4) = -\dfrac{5000}{25} = -200$. At month 4 the app is losing about 200 subscribers per month.

(b) $N(4) = 1000$. $N(4.5) \approx 1000 - 200 \cdot 0.5 = 900$ subscribers. (Exact: 909.09.)

4. (a) $f(3) = 2$, $f'(3) = -1$, $f''(3) = 0$, $f'''(3) = 3! \cdot \dfrac{1}{2} = 3$.

(b) Tangent: $L(x) = 2 - (x - 3)$, slope -1. $P_3 - P_1 = \dfrac{1}{2}(x - 3)^3$: odd power, so f crosses its tangent. Below it for $x < 3$ (concave down), above for $x > 3$ (concave up). Inflection point at $x = 3$.

5. $\Big[2x^{3/2} - 2\ln x\Big]_1^4 = (16 - 2\ln 4) - (2 - 0) = 14 - 2\ln 4 = 14 - 4\ln 2 \approx 11.23$.

6. Intersections: $x^2 = 2x \Rightarrow x = 0$ or $x = 2$. On $[0, 2]$ the line is on top ($2x \geq x^2$).
Area $= \displaystyle\int_0^2 (2x - x^2)\,dx = \Big[x^2 - \frac{x^3}{3}\Big]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.

7. (a) $P'(t) = 6 - 2t = 0 \Rightarrow t = 3$, $P'' = -2 < 0$. $P(3) = 21$; endpoints $P(0) = 12$, $P(6) = 12$. Highest profit after 3 hours: €21,000.

(b) $\displaystyle\frac{1}{6}\int_0^6 (12 + 6t - t^2)\,dt = \frac{1}{6}\Big[12t + 3t^2 - \frac{t^3}{3}\Big]_0^6 = \frac{108}{6} = 18$ thousand €.

Good luck on Wednesday.